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PipeSD

An Efficient Cloud-Edge Collaborative Pipeline Inference Framework with Speculative Decoding

Jiale Wang

Paper: https://arxiv.org/pdf/2605.13319 Code: https://github.com/Nuyoahwjl/PipeSD

Motivation

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Why Conventional Cloud-Edge Speculative Decoding Is Still Slow

The edge draft model generates a block of draft tokens. → *The block is uploaded only after generation is complete.* → *The cloud target model performs Non-Autoregressive Verification, or NAV.* → *The edge waits for the verification result before starting the next round.*

Two Main Bottlenecks

Serial generation and communication: *the edge processor is idle during upload, while the network is idle during draft generation.*

Inflexible fixed-length or single-threshold NAV triggering: *triggering too early repeatedly pays communication startup cost, while triggering too late accumulates low-quality tokens and increases rollback.*

NAV (Non-Autoregressive Verification)

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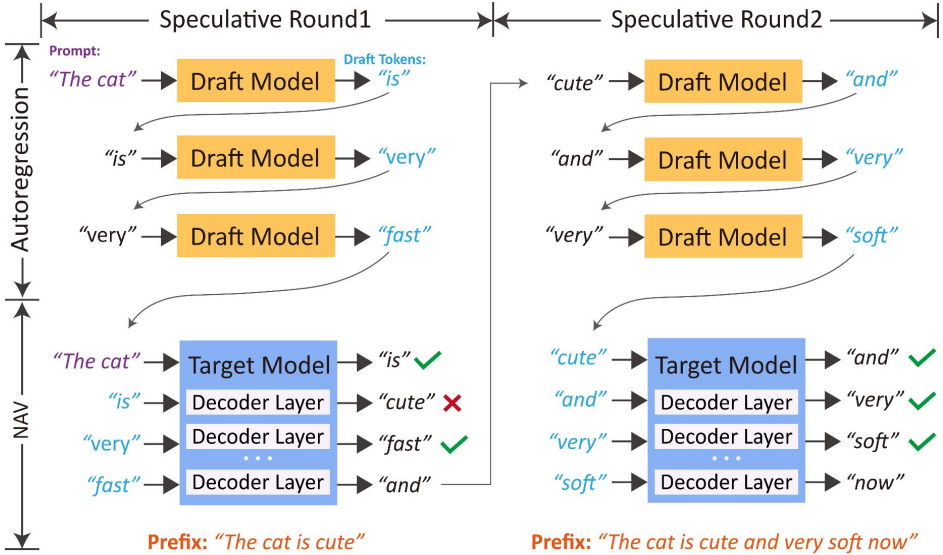
Let the edge draft model generate the candidate sequence $D=(d_1,d_2, \ldots ,d_{\square})$.

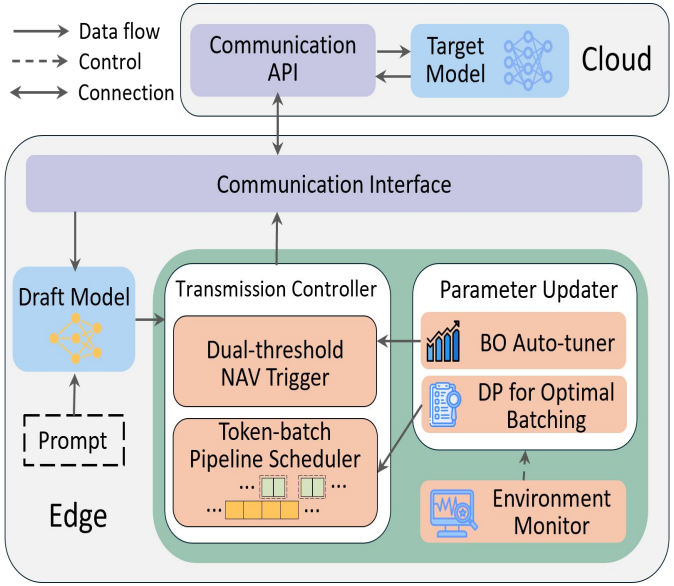
The cloud target model checks this sequence with one parallel forward pass:

- *It accepts the longest prefix that satisfies the target-model verification rule.*
- *It produces one additional target token after the accepted prefix.*
- *If a rejection occurs, all draft tokens after the rejection point become invalid.*
- *The context of the next round must be based on the cloud-confirmed sequence.*



Critical Reuse Condition: *Edge-side work generated in advance can be reused only if the target model accepts the entire parent draft and its additional target token matches the token assumed by the proactive branch.*

PipeSD Architecture



Communication API: *receives candidate batches and their round metadata.*
Target Model: *performs NAV and returns the accepted prefix plus one additional target token.*

Draft Model: *autoregressively produces candidate tokens.*
Transmission Controller: *combines the **DP** batch scheduler and the dual-threshold NAV trigger.*
Environment Monitor: *estimates token-generation time (γ), communication startup cost (α), and per-token communication time (β).*
Parameter Updater: *uses **Bayesian optimization** to update the thresholds and recomputes DP batch boundaries when the environment changes.*
Communication Interface: *uploads batches and receives NAV results.*

Pipeline Workflow

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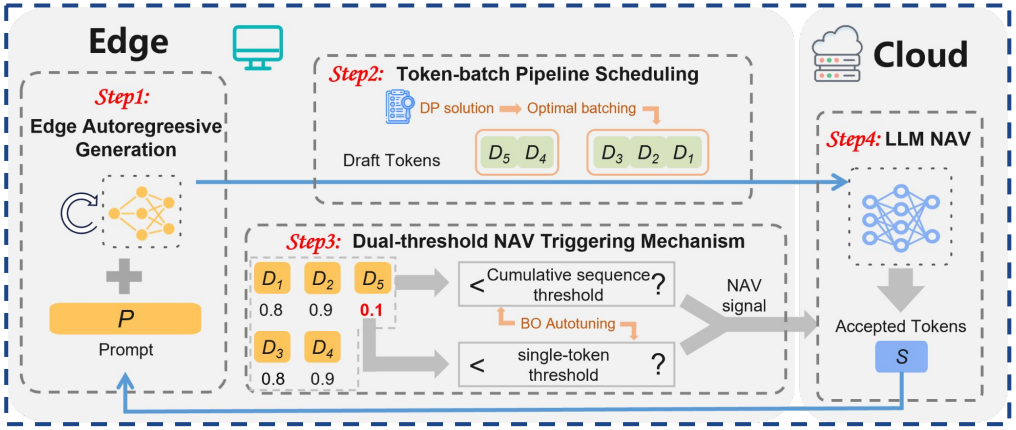
1. **Draft Generation:** *the edge generates tokens one by one and records their confidence.*
2. **Token-Batch Scheduling:** *once the current DP boundary is reached, the completed batch is uploaded asynchronously while generation continues.*
3. **Dual-Threshold Triggering:** *after every token, the system checks both token-level and cumulative confidence; crossing either threshold triggers NAV.*
4. **Cloud NAV:** *the cloud verifies the uploaded candidates and returns accepted tokens plus one additional target token.*
5. **Speculative Continuation:** *while waiting for NAV, the edge generates and may upload next-round candidates into an isolated proactive buffer.*
6. **Commit or Discard:** *proactive data is reused only when the parent draft is fully accepted and the additional token matches; otherwise it is cancelled and discarded.*



DP Scheduling

The paper models the transmission time and the generation time of a batch of size (b) as:

$$T_{comm}(b)=\alpha+\beta b,\ T_{gen}(b)=\gamma b.$$

For a predicted window of N tokens, select batch boundaries that minimize the completion time of the final transmission.

$$dp[j]=\min_{0\le i<j}\left[\max(dp[i],\gamma j)+\alpha+\beta(j-i)\right]$$

- *A batch can be sent only after all tokens in that batch have been generated and the previous transmission has completed.*
- *Dynamic programming enumerates the previous partition point with complexity ($O(N^2)$).*
- *($\hat{N}$) starts at 20 and is later updated using the moving average of draft lengths from the most recent 100 rounds.*
- *If NAV returns early, the active schedule is interrupted and unsent tokens are cleared.*

BO Threshold

For the (n)-th draft token:

- Token-level confidence: $P(d_n)$
- Cumulative sequence confidence: $C_n = \prod_{i=1}^{n} P(d_i)$

NAV is triggered when: $C_n \leq R_1 \text{ or } P(d_n) \leq R_2$.

Bayesian optimization searches for thresholds that minimize average time per token (TPT).

$$(R_1^*, R_2^*) = \underset{(R_1,R_2)\in(0,1)^2}{\operatorname{argmin}} \ \mathrm{TPT}(R_1, R_2)$$

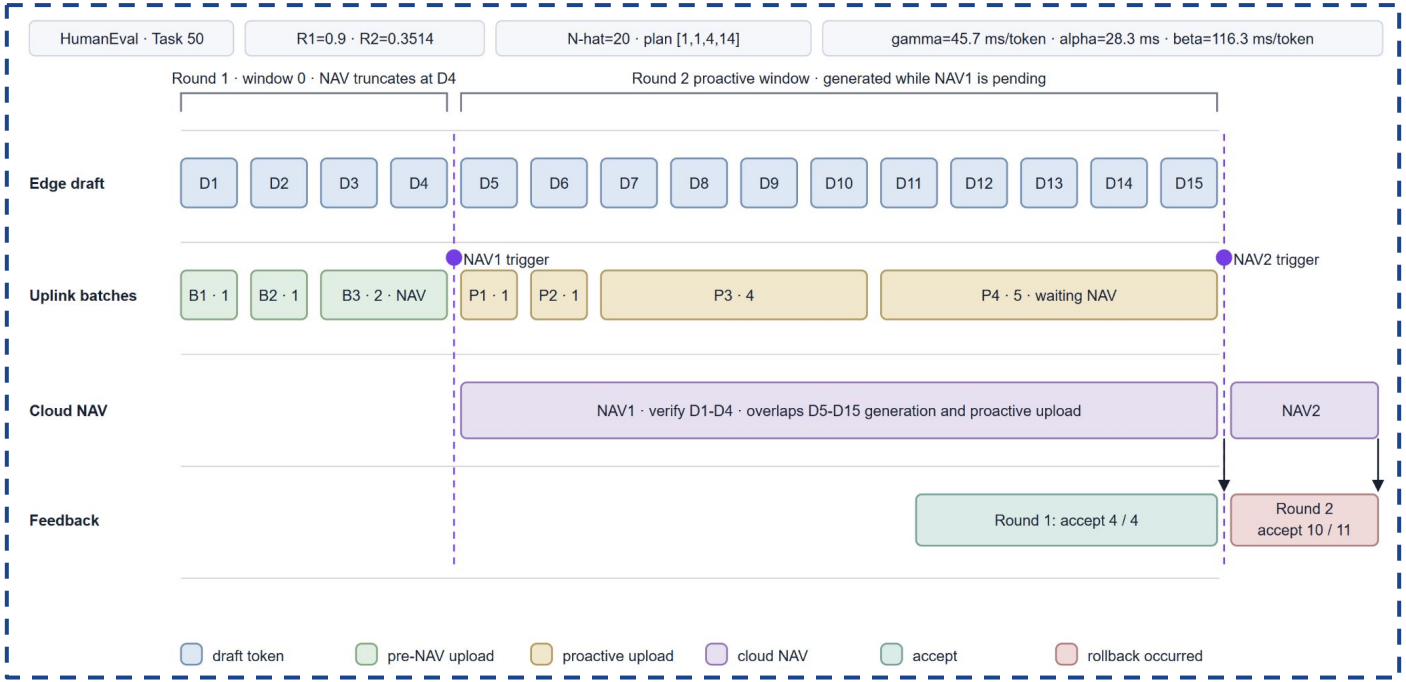
- *The token threshold detects a sudden local confidence drop.*
- *The cumulative threshold detects the growing risk created by several individually plausible tokens.*
- *The current implementation follows 16 budget and uses Expected Improvement with `xi=0.1`.*
- *PipeSD runs BO once at startup to initialize (R1,R2), then asynchronously re-optimizes the thresholds when performance changes significantly, enabling online adaptation.*

Data Lifecycle

Cloud Verification Result	Required Action
Only part of the parent draft is accepted	*Roll back tokens after the rejection point and cancel or discard next-round proactive data*
The full draft is accepted, but the additional target token does not match	*The proactive context is incorrect; discard it and restart from the target token*
The full draft is accepted and the additional token matches	*Promote the isolated proactive buffer to the next formal round*

Once proactive data is deemed invalid, its cleanup depends on its transmission stage. Tokens still stored locally are discarded immediately, queued batches are cancelled on a best-effort basis, while in-flight or cloud-buffered batches can no longer be retracted and must be discarded through cloud-side parent-round validation.

An Example


HumanEval · Task 50
R1=0.9 · R2=0.3514
N-hat=20 · plan [1,1,4,14]
gamma=45.7 ms/token · alpha=28.3 ms · beta=116.3 ms/token
Round 1 · window 0 · NAV truncates at D4
Round 2 proactive window · generated while NAV1 is pending
Edge draft
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
NAV1 trigger
NAV2 trigger
Uplink batches
B1 · 1
B2 · 1
B3 · 2 · NAV
P1 · 1
P2 · 1
P3 · 4
P4 · 5 · waiting NAV
Cloud NAV
NAV1 · verify D1-D4 · overlaps D5-D15 generation and proactive upload
NAV2
Feedback
Round 1: accept 4 / 4
Round 2
accept 10 / 11
draft token
pre-NAV upload
proactive upload
cloud NAV
accept
rollback occurred

Reproduction Work

Role	Reproduction Setup	Model
Cloud	NVIDIA RTX PRO 6000	HumanEval: DeepSeek-Coder-6.7B GSM8K: Llama-2-7B-Chat
Edge	CPU (`n_gpu_layers=0`)	HumanEval: DeepSeek-Coder-1.3B GSM8K: TinyLlama-1.1B-Chat-v1.0

Table 1 Scenario 1

- **Network:** *SoftwareLink emulates the paper's 20/200 Mbps link as 2.5 MB/s uplink and 25 MB/s downlink, while retaining real loopback HTTP and target-compute latency.*
- **Link Contention:** *Uplink and downlink use independent FIFOs; primary and proactive uploads share one uplink queue and therefore compete for the same 20 Mbps resource.*
- **DP Scheduling:** *Initial values are a=25 ms, B=0.29/2.5=0.116 s/token, γ=0.036 s/token, and N=20. Runtime estimates trigger batch replanning when they change by more than 20%.*
- ***BO**: R1/R2 are searched once before formal evaluation and remain fixed during inference; online BO threshold updates are not yet implemented.*
- **Evaluation:** *Seed 3407, at most 128 generated tokens per sample, and exactly 1,000 cloud-accepted tokens per method. TPT is total end-to-end time divided by accepted tokens; NVML samples GPU power every 5 ms for prompt prefill and target NAV only.*

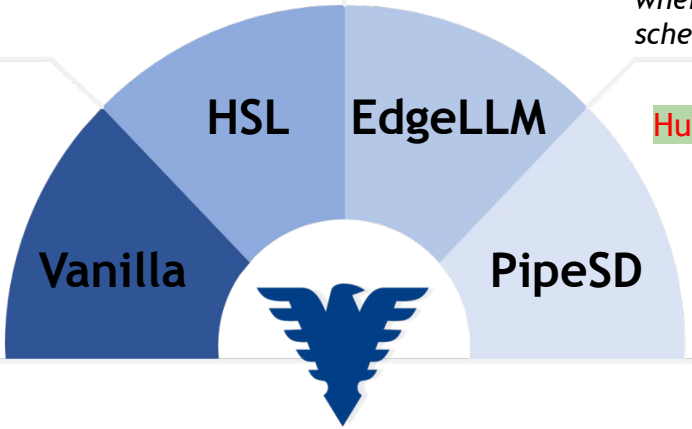
Baselines

HumanEval:0.95 GSM8K:0.7

Triggers NAV when a token's confidence falls below a threshold, uploads the full block and pauses generation while waiting.

HumanEval:6 GSM8K:4

Generates a fixed number of draft tokens, uploads the whole block, and waits for NAV without overlap.



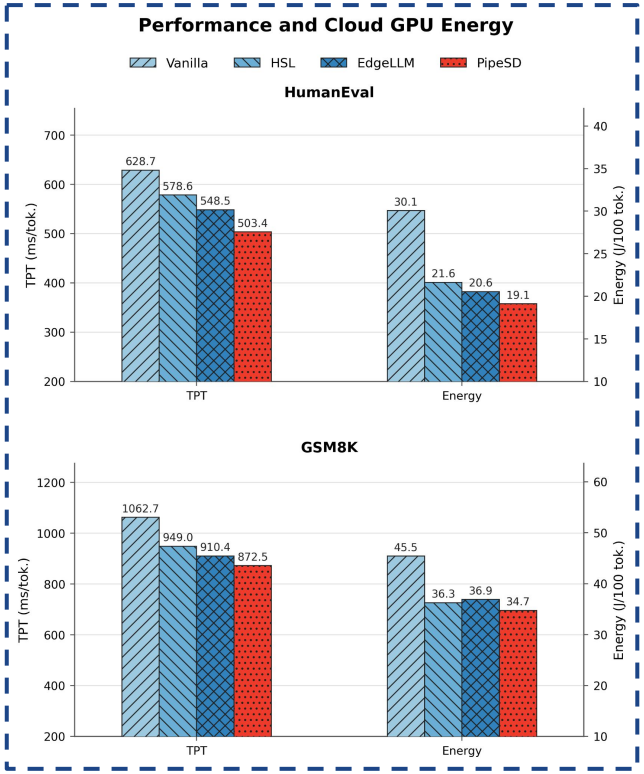
HumanEval:0.92 GSM8K:0.5

*Triggers NAV using cumulative sequence confidence, continues drafting during verification, and multiplies the threshold by **0.8** when accepted tokens reach the scheduling window .*

HumanEval:(0.9,0.35) GSM8K:(0.64,0.4)

Combines token and sequence-level thresholds with DP-based batch pipelining, continuing generation during NAV with safe reuse or rollback.

Evaluation Results



Task / Method	TPT (ms/ token)	Energy (J/100 tokens)
HumanEval Vanilla	628.705	30.060
HumanEval HSL	578.558	21.643
HumanEval EdgeLLM	548.539	20.552
HumanEval PipeSD	503.444	19.121
GSM8K Vanilla	1062.706	45.453
GSM8K HSL	949.016	36.254
GSM8K EdgeLLM	910.411	36.900
GSM8K PipeSD	872.472	34.743

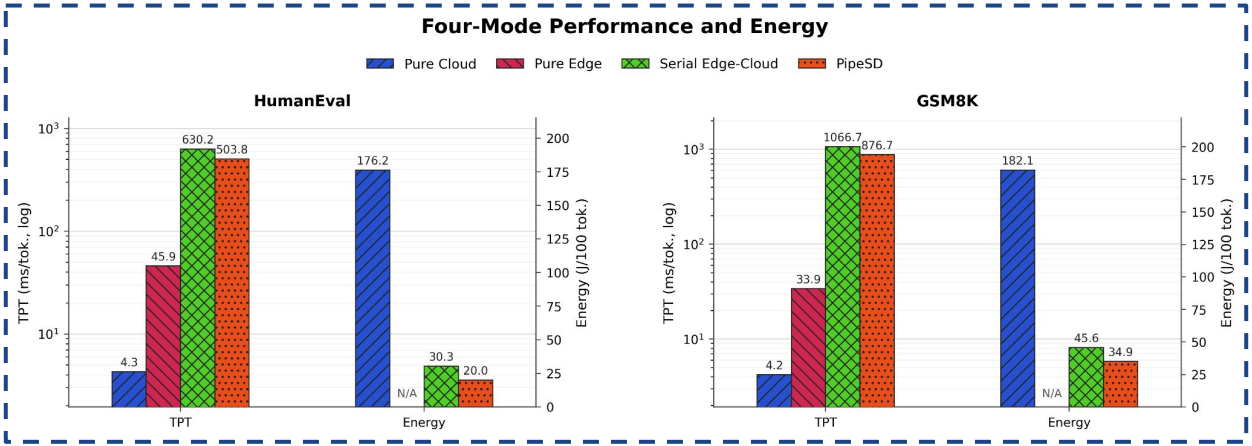
Evaluation Results

Task / Method	Draft Length	Draft Tokens	Accept Rate	Total NAVs	Rollback Rate	Discard Rate
HumanEval Vanilla	5.881	1388	72.0%	236	42.8%	–
HumanEval HSL	4.408	1058	94.5%	240	20.8%	–
HumanEval EdgeLLM	5.841	1250	80.0%	214	31.3%	46.9%
HumanEval PipeSD	5.279	1077	92.9%	204	24.0%	25.9%
GSM8K Vanilla	3.797	1815	55.1%	478	60.9%	–
GSM8K HSL	2.998	1361	73.5%	454	46.5%	–
GSM8K EdgeLLM	2.708	1263	79.2%	466	36.1%	50.0%
GSM8K PipeSD	2.995	1261	79.3%	421	46.3%	53.1%

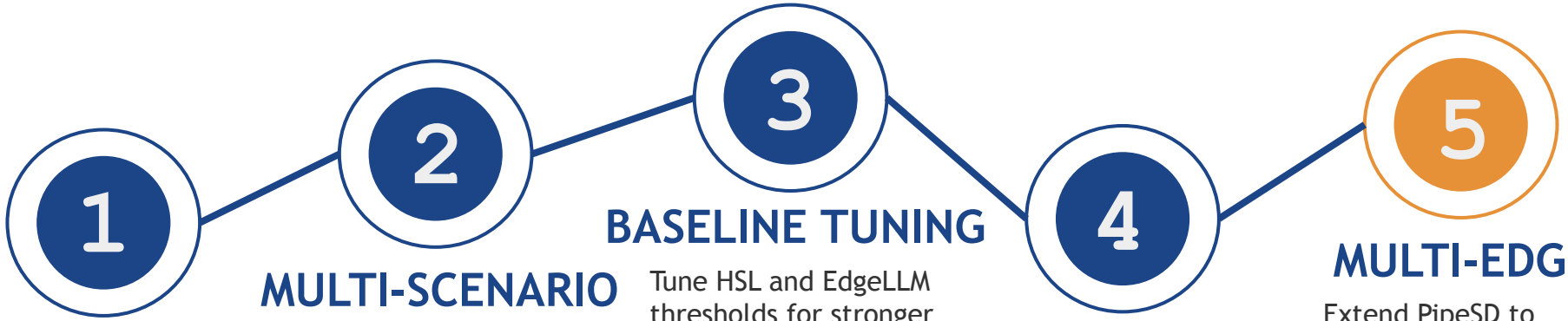
Rollback Rate is the number of NAV calls that reject at least one draft token divided by the total NAV count. It measures the fraction of NAV rounds with a rollback, not the fraction of rejected tokens.
Discard Rate = Discard / (Promoted + Discard), where Promoted denotes proactive tokens promoted to the next-round candidate, while Discard denotes proactive tokens invalidated by parent verification failure or an extra-token mismatch.

Four Modes

Pure Cloud: *The target model performs prompt prefill and full autoregressive decoding on one GPU without network transfer or NAV, and energy covers only this GPU computation.*
Pure Edge: *The draft model performs standalone autoregressive decoding on the CPU without network, cloud, or NAV, while energy is unavailable due to missing RAPL access.*
Serial Edge-Cloud SD: *The CPU draft generates and uploads each fixed window before the GPU target performs NAV sequentially, with energy covering cloud prefill and active NAV computation.*
PipeSD: *PipeSD uses the same draft/target models and link, overlaps generation with DP-batched transfer, and continues drafting during NAV waits, with energy likewise covering cloud prefill and active NAV computation.*



Plan



ONLINE BO

Asynchronously update scheduling thresholds during inference.

MULTI-SCENARIO

Evaluate diverse bandwidths, latencies, and network dynamics.

BASELINE TUNING

Tune HSL and EdgeLLM thresholds for stronger and fairer baselines.

MORE SAMPLES

Increase the evaluation sample size when time and compute permit.

MULTI-EDGE

Extend PipeSD to multiple edge devices sharing one cloud and study resource-aware scheduling.

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Thank you for listening